

A Product-Kernel Weighted Graph for Flood-Rescue Event Clustering and Cluster-Level Priority Scoring

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Abstract. Flood-rescue coordinators must consolidate noisy, multi-modal reports into spatially actionable units before assigning scarce teams. We study a weighted graph whose edge weight is a product of a Gaussian geographic kernel and a temporal-contextual kernel, followed by graph sparsification, community detection, and cluster-level priority scoring. Our main result is analytic: for pruning threshold $0 < \theta < 1$, every retained product-kernel edge is shorter than $\sigma\sqrt{2\ln(1/\theta)}$; therefore a connected cluster with a shortest spanning path of at most h edges has diameter below $h\sigma\sqrt{2\ln(1/\theta)}$. An additive counterpart has no distance-independent guarantee. We evaluate the pipeline on a 485-report synthetic benchmark designed with spatially overlapping events, unequal densities, temporal overlap, multimodal labels, and mixed fake reports. Coordinate-only baselines reach ARI 0.630, whereas removing context lowers the proposed pipeline’s ARI by 0.174. Across 20 regenerated geometries, product gating obtains ARI 0.962 ± 0.013 versus 0.935 ± 0.018 for additive weighting at the shared operational default; the paired 95% confidence interval of the difference excludes zero. Per-form threshold calibration narrows that empirical claim, so we attribute the contribution to the guarantee rather than universal superiority. Confidence scoring remains a weak heuristic, and real-data validation remains open.

Keywords: Flood rescue · Product kernel · Weighted graph · Community detection · Priority scoring.

1 Introduction

Disaster-response streams contain repeated, incomplete, and sometimes false reports. Treating each report as an independent rescue destination wastes travel capacity; grouping reports solely by coordinates can merge distinct physical events that overlap in space or split one event observed at several sites. Coordinators also need an ordered dispatch list, not clustering alone.

We investigate a pipeline that converts each report into compact attributes, constructs a spatial-temporal-contextual graph, detects communities, and ranks the resulting clusters. The central design is a *product kernel*: geography multiplies the temporal-contextual term instead of being added to it. Product kernels are established technology—bilateral filtering uses the same structural idea [17]—so the kernel form itself is not claimed as novel. Our contribution is its operational localization guarantee under threshold sparsification, together with a reproducible evaluation that separates that guarantee from threshold-calibration artifacts.

The contributions are:

- a localization lemma that bounds every retained edge and, consequently, the diameter of every connected cluster by quantities known before seeing the data; an additive weight has no corresponding distance-independent guarantee;
- a cluster priority function with an explicit vulnerability-amplification cap and a static population reference, evaluated with a multi-seed, multi-depot dispatch simulation; and
- an end-to-end evaluation on a synthetic benchmark whose difficulty is checked before comparing methods: coordinate-only baselines are weak, context ablation changes the partition, and single confidence features do not separate fake reports.

2 Related Work

Crisis sensing and edge deployment. CrisisMMD [1] and FloodNet [13] support multimodal disaster analysis. Lightweight vision and language models such as MobileNetV3 [9] and DistilBERT [16] motivate on-device extraction, while EmergencyNet [11] illustrates embedded inference. Our experiments do not benchmark these models: flood level and urgency are provided by the generator. Edge extraction is therefore deployment context, not an empirical contribution of this paper.

Graph clustering and spatial constraints. Louvain [4] and Leiden [18] optimize weighted modularity without a preset cluster count. K-Means [12], DBSCAN [7], HDBSCAN [5], and spectral clustering [20] provide complementary baselines. Spatial clustering is also addressed explicitly by ClustGeo [6], contiguity-constrained regionalization [2], and broader spatially constrained methods [21]. These lines make clear that neither multiplying kernels nor enforcing compact geography is new in isolation.

Product kernels and positioning. Bilateral filtering multiplies spatial and range kernels so either factor can veto an interaction [17]. We transpose that mechanism from pixels to rescue reports. The additive form used below is a controlled ablation constructed by us, not a claimed representation of all prior spatial clustering. Table 1 locates the narrower contribution.

Table 1. Positioning of the evaluated contribution.

Line of work	Product form	Spatial constraint	Cluster ranking	Bound
Bilateral filtering [17]	✓	local	–	–
ClustGeo/regionalization [6,2]	–	✓	–	–
Graph event detection [15,3]	~	~	–	–
This work	✓	✓	✓	✓

Priority and equity. Multi-criteria emergency allocation has long treated objectives and weights as policy choices [14,19]. Humanitarian logistics also exposes genuine tension among equity, efficiency, and efficacy [8,10]. Our vulnerability multiplier is an explicit, bounded policy control; outcome evaluation is required because merely changing a ranking does not demonstrate benefit.

3 Method

3.1 Report Attributes and Deployment Scope

Each report is represented as

$$v_i = (L_i, T_i, F_i, E_i, N_i, V_i, C_i),$$

where L_i and T_i are location and time, $F_i, E_i \in [0, 1]$ are flood level and urgency, N_i is reported population, $V_i \geq 0$ is vulnerability mass, and $C_i \in (0, 1)$ is confidence. A compact JSON representation occupies 105–111 bytes in our serialization experiment. A prospective mobile client can extract F_i and E_i using lightweight image/text models and send only these attributes. Because no device latency, memory, energy, or model-accuracy result is reported here, this remains a proposed deployment architecture.

Confidence is the interpretable heuristic

$$C_i = \text{sigmoid}(b_0 + b_1 \mathbf{1}[\text{image}] + b_2 \log(1 + n_{\text{corrob}, i})).$$

It downweights inputs to the priority score; it is not treated as a validated misinformation detector.

3.2 Product-Kernel Graph and Localization Guarantee

For reports i, j , let d_{ij} be Haversine distance and Δt_{ij} the absolute time difference. We define

$$S_{\text{geo}}(d) = \exp[-d^2/(2\sigma^2)], \quad (1)$$

$$S_{\text{temp}}(\Delta t) = \exp[-\Delta t/\tau_t], \quad (2)$$

$$S_{\text{ctx}}(i, j) = \exp[-|F_i - F_j|/\tau_F - |E_i - E_j|/\tau_E], \quad (3)$$

$$w_{ij}^\times = S_{\text{geo}}(d_{ij})[\beta S_{\text{temp}} + \gamma S_{\text{ctx}}], \quad (4)$$

with $\beta, \gamma \geq 0$ and $\beta + \gamma \leq 1$. We retain weights above θ and then keep the union of each node's k strongest edges. The k -nearest operation can only delete edges, so the following result survives both sparsification stages.

Lemma 1 (Localization of product-kernel edges). *For $0 < \theta < 1$, every retained edge under (4) satisfies*

$$d_{ij} < r_\theta := \sigma \sqrt{2 \ln(1/\theta)}.$$

If any two vertices in a connected component are joined by a path of at most h edges, that component has geographic diameter below hr_θ .

Proof. Since both similarities are at most one and $\beta + \gamma \leq 1$, $w_{ij}^\times \leq S_{\text{geo}}(d_{ij})$. Thus $w_{ij}^\times > \theta$ implies $\exp[-d_{ij}^2/(2\sigma^2)] > \theta$, which is equivalent to $d_{ij} < \sigma \sqrt{2 \ln(1/\theta)}$. Applying the triangle inequality along a path of at most h retained edges gives the component bound.

For $\sigma = 700$ m and $\theta = 0.05$, every retained edge is shorter than 1.713 km. This also enables safe spatial blocking: pairs outside r_θ need not be evaluated when constructing the graph.

Why addition has no analogous guarantee. The controlled baseline is

$$w_{ij}^+ = \alpha S_{\text{geo}} + \beta S_{\text{temp}} + \gamma S_{\text{ctx}}.$$

Its non-geographic terms create a positive floor independent of distance. Because $F, E \in [0, 1]$, for example, $\gamma S_{\text{ctx}} \geq \gamma \exp(-1/\tau_F - 1/\tau_E) > 0$ when $\gamma > 0$. For any threshold below that floor, equally timed or context-compatible reports can remain connected at arbitrary geographic separation. This does not say additive weighting cannot form compact clusters after calibration; it says compactness is not guaranteed by θ and σ alone.

3.3 Community Detection and Cluster Priority

Louvain partitions the sparse weighted graph; Leiden is checked as a drop-in alternative. For cluster C_k , we compute

$$\tilde{E}_k = |C_k|^{-1} \sum_{i \in C_k} E_i C_i, \quad \tilde{F}_k = \max_{i \in C_k} (F_i C_i), \quad (5)$$

$$\tilde{N}_k = \min \left\{ 1, \frac{\log(1 + \sum_{i \in C_k} N_i C_i)}{\log(1 + N_{\text{ref}})} \right\}, \quad A_k = 1 + (\mu - 1) \tanh \left(\frac{\sum_{i \in C_k} V_i}{s} \right), \quad (6)$$

$$R_k = \omega_E \tilde{E}_k + \omega_F \tilde{F}_k + \omega_N \tilde{N}_k, \quad \mathcal{P}(C_k) = A_k R_k. \quad (7)$$

Weights ω sum to one and $\mu \in [1, 2]$, so $\mathcal{P} \in [0, 2)$. We use a static $N_{\text{ref}} = 500$ rather than the largest current cluster. With a dynamic reference, an unchanged cluster's score can change when an unrelated larger cluster appears.

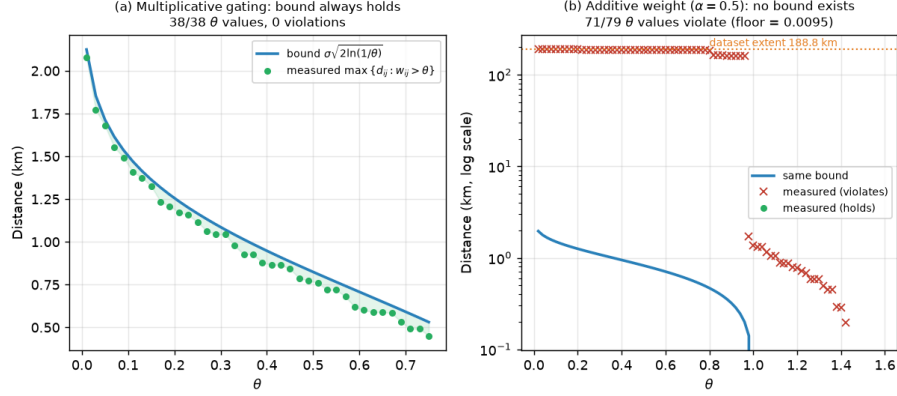


Fig. 1. Measured longest retained edge versus Lemma 1. Product gating has zero violations in 38 thresholds. The additive ablation ($\alpha = 0.5$) violates the same bound at 71 of 79 thresholds; the bound is not claimed for that form.

Table 2. Dataset-hardness gates (seed 42).

Gate	Result	Required
K-Means / Haversine agglomerative, coordinates only (ARI)	0.630 / 0.412	both < 0.75
Remove S_{ctx} : ARI drop	0.174	≥ 0.08
Remove S_{ctx} : drop on overlapping pairs	0.396	diagnostic
Sweep (τ_F, τ_E) : ARI range	0.071	≥ 0.01
Strongest single fake-report feature (AUC)	0.690	≤ 0.75
Fake reports placed inside event regions	66.7%	$\geq 55\%$

4 Experimental Design

4.1 Hard Synthetic Benchmark

The seed-42 dataset contains 485 reports: 421 labeled observations from 13 physical events, 60 noise reports, and a four-report fake campaign. Event sizes and spreads vary from 8–70 observations and 120–900 m. Three event pairs have overlapping spatial envelopes but contrasting (F, E) profiles; another pair shares a location but is separated by 3.5 h; one label is bimodal. Thirty-nine reports are fake, 66.7% of them are placed inside event regions, and image availability overlaps between real and fake reports.

Table 2 is evaluated before method comparison. All five pre-registered gates pass. In particular, the combined product graph is not guaranteed to win by coordinates alone, and the context term changes the partition materially.

4.2 Parameters, Metrics, and Statistical Protocol

Defaults are $\sigma = 700$ m, $\tau_t = 45$ min, $(\tau_F, \tau_E) = (0.25, 0.35)$, $(\beta, \gamma) = (0.5, 0.5)$, $\theta = 0.05$, $k = 12$, resolution 1.0, $(\omega_E, \omega_F, \omega_N) = (0.34, 0.33, 0.33)$, $s = 10$, and

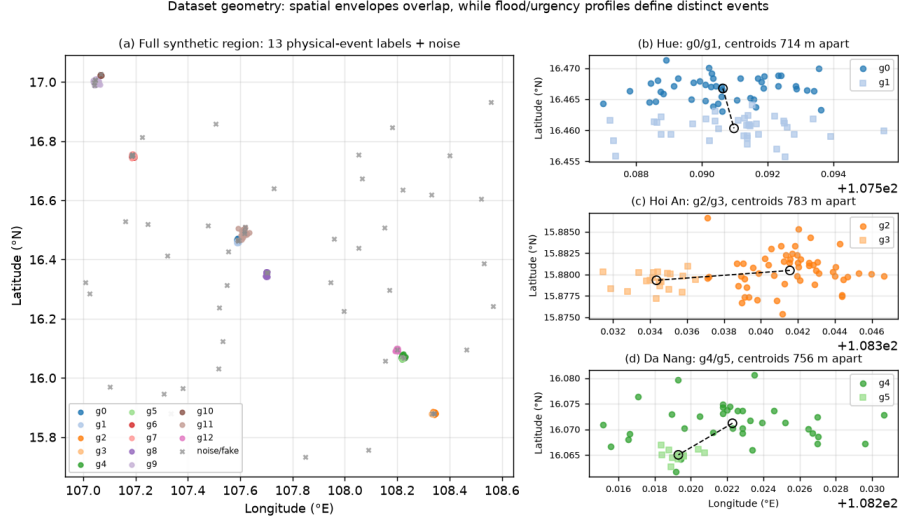


Fig. 2. Dataset distribution and the three spatially overlapping event pairs. Their centroids are 714–783 m apart and their spatial envelopes overlap; the labels are defined by physical event context rather than by an island center.

$\mu = 2$. We report ARI/NMI only on labeled points. The primary geographic statistics are computed identically for every method over predicted clusters containing at least one labeled point; noise-only clusters and unassigned points are reported separately. This prevents a method’s noise-bin convention from determining its apparent diameter.

Twenty-seed tests regenerate inter-event geometry, group sizes, and spreads (not only point jitter). Paired differences use bootstrap 95% confidence intervals and Wilcoxon signed-rank tests. A result is called different only if the paired interval excludes zero.

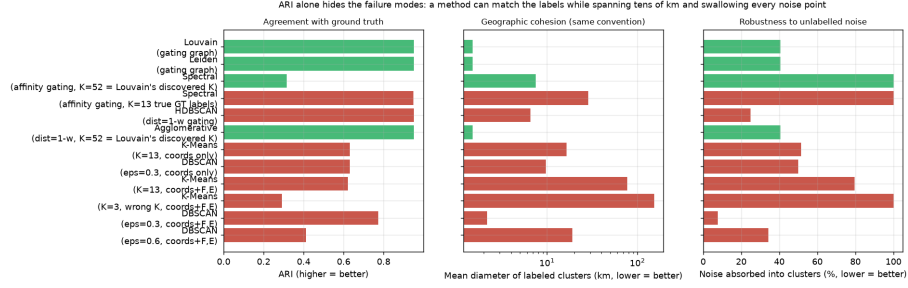
5 Results

5.1 Clustering Across Regenerated Geometries

At the shared operational default $\theta = 0.05$, product gating has mean ARI 0.9616 ± 0.0134 (95% CI $[0.9562, 0.9675]$), versus 0.9348 ± 0.0179 ($[0.9272, 0.9426]$) for additive $\alpha = 1$. The paired ARI difference is 0.0268, CI $[0.0198, 0.0345]$, $p = 1.32 \times 10^{-4}$. Mean labeled-cluster diameter is 1.54 km versus 98.53 km. These numbers characterize the shared default, not intrinsic superiority under every calibrated threshold.

Table 3. Selected 20-seed results at the shared default. CI is for the mean; Δ is product minus additive.

Metric	Product	Additive $\alpha = 1$	paired Δ CI
ARI	0.962 [0.956,0.968]	0.935 [0.927,0.943]	[0.020,0.035]
NMI	0.978 [0.976,0.981]	0.963 [0.960,0.967]	[0.011,0.019]
Mean labeled diameter (km)	1.54 [1.51,1.58]	98.53 [93.10,103.51]	[−101.97, −91.56]
Max labeled diameter (km)	4.44 [4.16,4.75]	192.27 [186.87,197.15]	[−192.65, −182.57]

**Fig. 3.** Baselines under a common geographic convention. ARI, labeled-cluster diameter, and noise absorption reveal different failure modes.

5.2 Baselines and Measurement Convention

On seed 42, Louvain and Leiden coincide (ARI 0.9515, mean labeled diameter 1.50 km). Agglomerative clustering on the same distance ties them but is given $K = 52$, the count discovered by Louvain. HDBSCAN reaches nearly the same ARI (0.9508), but its 15 labeled clusters average 6.58 km and reach 79.86 km; three additional clusters contain only noise. Coordinate-only K-Means reaches ARI 0.6304. Thus the evidence supports the weighted graph’s partition and automatic K , not a universal claim that Louvain is superior.

5.3 Context and Parameter Sensitivity

Removing context lowers global ARI from 0.9515 to 0.7775 and overlapping-pair ARI from 0.9379 to 0.5419. The priority rankings remain similar ($\tau = 0.9774$), so context primarily affects event separation rather than re-ranking unchanged locations. The wider context sweep is non-flat (ARI 0.8809–0.9515), while σ and resolution sweeps span 0.182 and 0.132 ARI respectively.

5.4 Threshold Calibration and Lemma Verification

A shared numerical threshold is not a scale-invariant comparison: at $\theta = 0.05$, product gating retains 10.85% of pairs while additive variants retain 95.69–97.97%. Figure 6 is therefore a diagnostic of the default, not evidence that addition can never produce compact clusters.

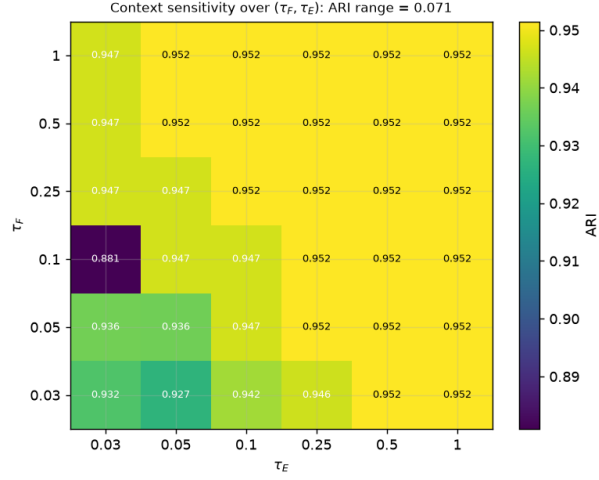


Fig. 4. Context sensitivity. The earlier narrow sweep lay on a plateau; the wider pre-registered grid exposes a 0.071 ARI range.

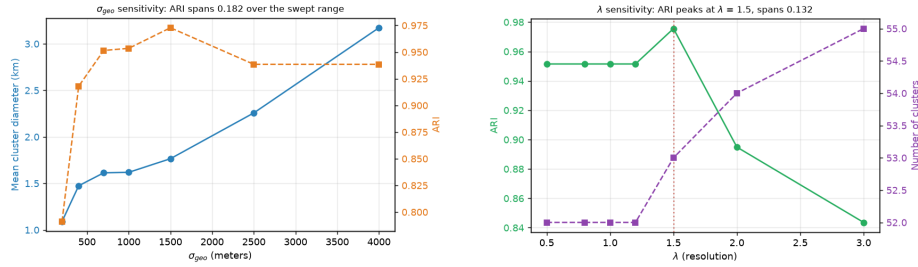


Fig. 5. Sensitivity to geographic scale (left) and modularity resolution (right). The pipeline is not parameter-invariant.

We calibrate each form separately using an operational criterion that does not use labels: maximum diameter below 5 km, fewer than half of events in singletons, and 5–40 multi-report clusters. Additive $\alpha = 1$ has 37 usable grid points versus 34 for product gating and reaches a 3.17 km maximum diameter at its best-ARI threshold. Consequently, there is no defensible “wider window” contribution. The invariant result is Lemma 1: zero measured violations for product gating, while all additive variants violate the analogous bound.

5.5 Priority Stability and Dispatch Outcomes

Perturbing ω by ± 0.10 yields mean Kendall $\tau = 0.9665$ (minimum 0.9276), but the top-three set is preserved in 87%, not 100%, of trials. Structural perturbations of σ yield matched-cluster $\tau \geq 0.989$. In an online stress sequence, the

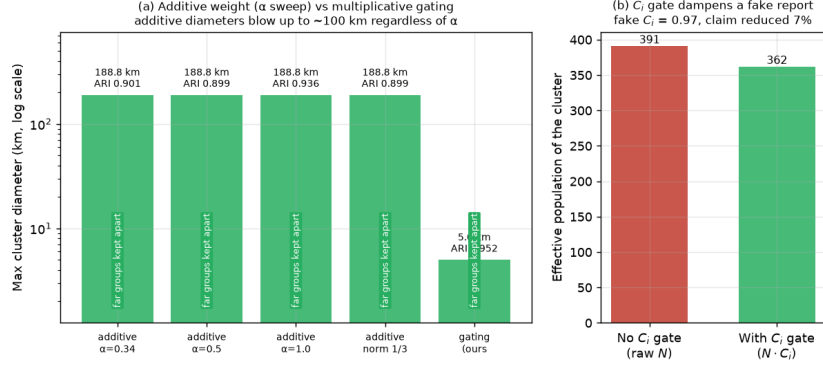


Fig. 6. Shared-default diagnostic. Additive weights form near-complete graphs at $\theta = 0.05$; their large diameters here must not be read as an intrinsic limit of addition. The right panel illustrates confidence weighting, not misinformation detection.

Table 4. Per-form calibration. Normalized windows use $\theta/w_{\max}$; retained windows are fractions of all possible edges.

Form	Best ARI	θ	Usable $\theta/w_{\max}$	Retained frac.	Viol.
Product	0.9731	0.39	[0.033,0.753]	[0.006,0.112]	0/38
Additive $\alpha = .5$	0.9768	0.90	[0.670,0.851]	[0.006,0.040]	71/79
Additive $\alpha = 1$	0.9652	1.34	[0.501,0.877]	[0.006,0.078]	95/99

static population reference keeps anchor-cluster priority unchanged ($\tau = 1$, zero score drift); a dynamic maximum ends at $\tau = 0.9111$ with maximum absolute priority drift 0.1265.

The dispatch simulation serves only clusters with at least two reports and one report with $C_i \geq 0.5$, uses three boats, three depot positions, and 20 regenerated geometries. Its primary, formula-external metric weights arrival time by vulnerable population only in clusters with $F_{\max} > 0.7$. Product vulnerability amplification improves on omitting vulnerability by 14.82 min (paired CI $[-26.68, -3.73]$, $p = 0.0266$). Against additive vulnerability, the difference is only -1.43 min with CI $[-10.91, 7.88]$ and $p = 0.756$. Therefore vulnerability-aware scoring helps relative to ignoring vulnerability, but the data do not establish that multiplication is better than addition.

5.6 Confidence Heuristic and Computational Cost

The combined fake score $1 - C_i$ has AUC 0.6919 ([0.6095,0.7693]) and AP 0.1546 ([0.1135,0.2267]) against random AP 0.0804. The single feature $-n_{\text{corrob}}$ has slightly lower AUC (0.6903) but higher AP (0.1680). Conditional AUC is 0.421, 0.468, and 0.623 for $n_{\text{corrob}} = 0, 1-5$, and > 5 , respectively. The four coordinated

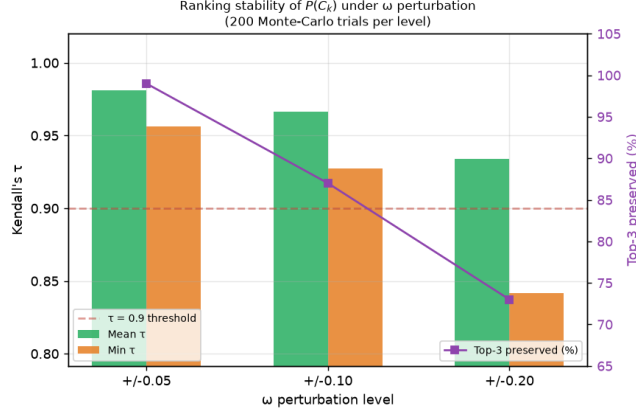


Fig. 7. Ranking stability under 200 weight perturbations per level. Kendall τ measures self-consistency, not correctness.

fake reports attain mean $C_i = 0.9273$, above the real-report mean 0.9035. Hence C_i is retained only as a transparent downweighting convention.

Constructing the dense weight matrix is $O(n^2)$; Louvain’s favorable sparse complexity does not change the end-to-end bound. In the recorded run, 7,200 events take 46.06 s on one CPU core, of which the vectorized matrix build takes 33.76 s. Lemma 1 suggests a scalable implementation: spatial indexing can omit all pairs outside r_θ before evaluating the other kernels.

6 Discussion and Threats to Validity

What is supported. The product form gives a closed-form, data-independent edge-localization guarantee and strong shared-default performance on a benchmark where coordinates alone do not solve the task. Context has a measurable role specifically on overlapping events. Static population normalization prevents unrelated score drift. These statements have direct analytic or multi-seed support.

What is not supported. Per-form calibration shows additive weighting can also produce compact, high-ARI partitions; the contribution is not a threshold-window ratio or universal empirical dominance. The confidence heuristic is not a reliable fake report detector. Product vulnerability amplification is not significantly better than additive vulnerability on the primary outcome. Edge-model accuracy and on-device resource use are not evaluated.

Validity limits. The benchmark is synthetic and covers one regional flood scenario. Its ground-truth physical events and hard cases are designed rather than observed. Twenty geometries quantify generator variability but cannot replace

field data. ARI/NMI score label agreement, whereas dispatch simulation depends on a simple greedy queue, fixed boat speed, and fixed service time. HDBSCAN and DBSCAN remain sensitive to their own hyperparameters, while Louvain depends on resolution, θ , and k . A real-data sanity check using geolocated, time-resolved flood reports is the highest-priority next step.

7 Conclusion

We presented a product-kernel weighted graph and a bounded cluster-priority score for flood-rescue coordination. The main contribution is a localization lemma: thresholded product edges, and therefore connected paths, have an explicit geographic bound controlled by σ and θ . A redesigned synthetic benchmark confirms that geography alone is insufficient and that context separates spatially overlapping events. Multi-seed results favor product gating at the shared operational default, while per-form calibration correctly narrows the empirical claim. Priority outcomes support including vulnerability but do not distinguish multiplicative from additive treatment; confidence scoring remains heuristic. Field validation and sparse, spatial-indexed graph construction are required before deployment.

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